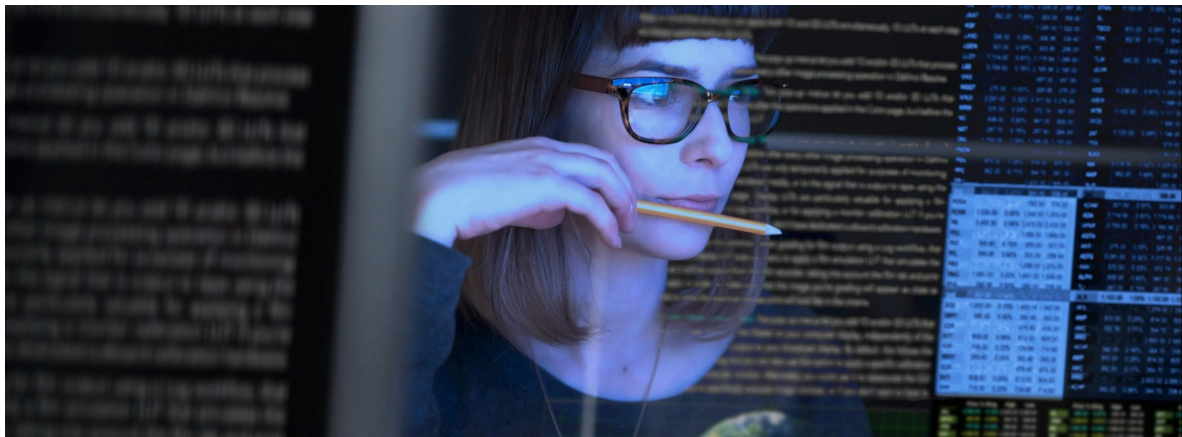




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Unit testing 1m



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When conducting testing in PHP, we can make use of the PHPUnit framework. This lets you create tests for functions, methods and classes. This means to effectively write unit tests, your program code would need to be written using PHP OOP. An alternate mixed unit test and UAT library is Codeception, which lets you not only write unit tests but also automated Acceptance and REST API tests.

The purpose of the unit test is to test just one program element at a time, which could involve several tests just for a single function. For example, you could have a function that accepts five numbers and returns an average as a result. You would require several individual tests to test each of the numbers, their types and the expected result for the numbers.

```
function Average(num1, num2, num3, num4, num5) {  
  
    return (num1 + num2 + num3 + num4 + num5) / 5;  
  
}
```

In the above function, we would have a test that calls Average with one or more numbers set, then compare the average. You may notice there are no range checks, checks for negative numbers, checks to make sure num1 . . . num5 are actual numbers and so on. The unit testing would check for this.

A rather overly simplistic unit test for the above function would be:

```
<?php  
  
assert(Average(1,1,1,1,1) == 5,"Incorrect average calculated");  
  
?>
```

The above would not cause an error as the average would be a one however the following would fail because five numbers are expected and we have supplied 1 where 1/5 is not equal to 1.

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<?php

```
assert(Average(1) == 1,"Expected five numbers");
```

```
?>
```

The above method of unit testing is the most basic, but least flexible, method of testing as it does not allow for testing of Class and methods, REST API, network calls and so on. It is ideal for testing individual functions and their expected return value based on the known inputs.

As you might have noticed, a unit test calls a function or method with one or more known values and expects a known response. This means when you want to make use of unit tests, you need to ensure that the code to be tested resides in the callable functions or methods.



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☒ Unit test

Non-Assessed

Write one or more PHP unit tests for your Resume PHP code. This means you need to put some of the code into various functions. Some expected functions:

- function dbconnect([credentials])
- function validatePOST([POST values])
- function saveMessage([check form data])

☐ I have completed this task

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